ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

# MECHATRONICS

## PAPER 1 · PRACTICE A

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 4. Section A: objective questions (30) · Section B: control systems and block diagrams, mechanical power transmission, system design and integration, fault-finding and maintenance strategies (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

## SECTION A

## QUESTION 1

**1.1** Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 In a closed-loop control system, the signal fed back to the controller comes from the …

A setpoint. B power supply. C actuator only. D sensor.

1.1.2 The steady-state error of a system is the …

A difference between the setpoint and the output after the system has settled. B time taken to reach the setpoint.
C highest value of the output. D noise in the signal.

1.1.3 In a compound gear train, the total gear ratio is found by …

A adding the numbers of teeth. B dividing the largest gear by the smallest.
C multiplying the ratios of each gear pair. D averaging the ratios.

1.1.4 Power transmitted by a rotating shaft is calculated with …

A $P = F \times s$. B $P = V \div I$. C $P = 2\pi nT \div 60$. D $P = m \times g$.

1.1.5 Maintenance done only after a machine has broken down is …

A breakdown (reactive) maintenance. B preventive maintenance.
C predictive maintenance. D planned shutdown.

1.1.6 Vibration analysis on a motor bearing is an example of …

A breakdown maintenance. B cleaning.
C calibration only. D predictive (condition-based) maintenance.

1.1.7 The time a system takes to go from 10% to 90% of its final value is the …

A settling time. B rise time. C dead time. D overshoot.

1.1.8 A ball screw is used to convert …

A linear motion into heat. B AC into DC.
C pressure into flow. D rotary motion into precise linear motion.

1.1.9 The efficiency of a machine is …

A input power ÷ output power.
B output power ÷ input power × 100.
C torque × speed.
D load ÷ effort.

1.1.10 A system that needs a person to adjust it when conditions change is a/an …

A closed-loop system.
B feedback system.
C open-loop system.
D servo system.

(10)

**1.2** Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A thermostat-controlled oven is an example of a closed-loop system.

1.2.2 Overshoot means the output goes past the setpoint before settling.

1.2.3 Gear trains with a high reduction ratio increase output speed.

1.2.4 Backlash is the small amount of free play between meshing gears.

1.2.5 Mean time between failures (MTBF) is a measure of reliability.

1.2.6 Calibration makes sure that a sensor's readings are accurate.

1.2.7 Couplings are used to join two shafts.

1.2.8 Fault-finding should start by replacing the most expensive part.

1.2.9 Thermal imaging can detect overheating electrical connections.

1.2.10 Increasing friction always increases the efficiency of a machine.

(10)

**1.3** Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

| | **COLUMN A** | | **COLUMN B** |
|---|---|---|---|
| 1.3.1 | Diagram showing the parts of a control system as boxes with signal arrows | A | summing junction (comparator) |
| 1.3.2 | Point where the feedback signal is subtracted from the setpoint | B | root cause analysis |
| 1.3.3 | Controller that switches fully on or fully off, e.g. a geyser thermostat | C | on/off (two-step) control |
| 1.3.4 | Range around the setpoint where an on/off controller does not switch | D | block diagram |
| 1.3.5 | Gear that changes direction of rotation without changing the ratio | E | idler gear |
| 1.3.6 | Gear set that gives very large reduction at 90° | F | MTTR |
| 1.3.7 | Average time taken to repair a fault | G | hysteresis (dead band) |
| 1.3.8 | Finding the basic cause of a repeated failure | H | worm drive |
| 1.3.9 | Record of all maintenance done on a machine | I | OEE |

| | | | |
|---|---|---|---|
| 1.3.10 | Measure combining availability, performance and quality of equipment | J | maintenance log (history card) |

(10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2: CONTROL SYSTEMS

**2.1** Draw a labelled block diagram of a closed-loop speed control system for a conveyor motor. Show the setpoint, summing junction, controller, actuator (motor drive and motor), process, sensor and feedback path. (8)

**2.2** Compare open-loop and closed-loop control under the headings accuracy, cost and response to disturbances. (6)

**2.3** Sketch a typical step response of a closed-loop system and indicate the rise time, overshoot, settling time and steady-state error. (6)

**[20]**

## QUESTION 3: MECHANICAL POWER TRANSMISSION

**3.1** In a compound gear train, gear A (15 teeth) on the motor shaft drives gear B (45 teeth). Gear C (20 teeth) is fixed on the same shaft as B and drives gear D (60 teeth). The motor turns at 1 800 r/min with a torque of 5 N·m.

3.1.1 Calculate the total gear ratio. (3)

3.1.2 Calculate the output speed of gear D. (2)

3.1.3 Calculate the input power. (3)

3.1.4 Calculate the output torque if the gear train is 90% efficient. (4)

**3.2** A conveyor belt is driven by a pulley with a diameter of 0,2 m at 120 r/min. Calculate the linear speed of the belt in m/s. (3)

**3.3** Explain what backlash is and why it is a problem in robot joints. Give ONE way to reduce it. (3)

**[18]**

## QUESTION 4: SYSTEM DESIGN AND INTEGRATION

A small dairy wants an automatic bottle-capping station. Bottles arrive on a conveyor, must be stopped under a capping head, capped with a pneumatic press, checked by a sensor and then released. The station must count bottles and stop if a cap is missing.

**4.1** List the sensors, actuators and the controller you would use for the station, with the purpose of each (at least SIX devices). (6)

**4.2** Write the sequence of operations for one capping cycle in numbered steps. (6)

**4.3** Name FOUR factors to consider when integrating the mechanical, electrical and control parts of the station. (4)

**[16]**

## QUESTION 5: FAULT-FINDING AND MAINTENANCE

**5.1** The capping press does not move, although the controller output LED is on. Describe a logical fault-finding procedure with at least FIVE steps. (5)

**5.2** Compare breakdown, preventive and predictive maintenance, giving ONE advantage and ONE disadvantage of each. (6)

**5.3** A machine operated for 720 hours in a month and had 4 breakdowns with a total repair time of 12 hours. Calculate the MTBF and the MTTR. (4)

**5.4** Explain why accurate maintenance records are important. (1)

**[16]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**